

ULIP Integration Requirement Document

Version No.: 2.0

Date: 30/03/2026

Project Name: ULIP

This document contains proprietary information of NLDSL. Unauthorized access, copying and replication are prohibited. This document must not be copied in whole or part by any means, without the written authorization of NLDSL, Noida, India.

Revision History

Version	Date	Author	Reviewer	Significant Changes
1.0	03/10/2024	Hasan	Nayab Ali	Baseline Version
2.0	30/03/2026	Gaurav Tyagi	Pranab Manna	Updated

Table of Contents

Introduction	4
1.1 Purpose	4
The Overall Description	4
1.2 Data Integration between TGVAHAN System and ULIP application	4
1.3 TGVAHAN/01	4
1.3.1 Technical Approach.....	5
1.4 Data Transmission & Authentication Mechanism	8

Introduction

1.1 Purpose

The purpose of this document is to define and provide details for accessing API for TGVAHAN system information from the ULIP system.

This document details the following points:

1. **Data Integration between TGVAHAN and ULIP web service:** Identification of operational points where data integration between the Telangana Transport Department and ULIP web service will be needed.
2. **Technical Approach** for Performing Data Integration

The Overall Description

The user will share information such as vehicle numbers based on data sent by the user. The ULIP system will call the respective **TGVAHAN** API to access information.

The following are the points for accessing cargo information: -

1. This API will share data based on information provided by the user.
2. ULIP will expose API for getting data from **TGVAHAN** API.
3. **TGVAHAN /01** API will take the vehicle number from the user and get data from TGVAHAN API.

1.2 Data Integration between TGVAHAN System and ULIP application

The user will share information with the ULIP system based on the data shared ULIP will provide data to the user.

1.3 TGVAHAN/01

This API will take the vehicle number and connect with the TGVAHAN API to get data.

S. No.	Field Name	Field Description	Format	Required/Not – Required	Length
1	vehiclenumber	Vehicle number	^[A-Z0-9]{5,11}\$	Required	Max 11

1.3.1 Technical Approach

1. All the data will be shared through the rest web services.
2. The ULIP system will provide the Vehicle number for which detail is required.
3. TGVAHAN API will share details of the vehicle registration information that exists in the TGVAHAN system in JSON format.
4. TGVAHAN Integration will add JSON in the **response** key in the response body.
5. The ULIP system will provide the data in JSON Object format in the response body.

Example:

Vehicle number TG08T1024 is shared by the ULIP system, and then TGVAHAN-integration will request TGVAHAN API.

Request:

The request must be of the following format, where the vehicle number is passed within the URL body in JSON format.

For staging –

<https://www.ulipstaging.dpiit.gov.in/ulip/v1.0.0/TGVAHAN/01>

```
{"vehiclenumber": "TG08T1024"}
```

Curl URL

```
curl --location --request POST 'https://www.ulipstaging.dpiit.gov.in/ulip/v1.0.0/TGVAHAN/01' \
--header 'authorization: Bearer eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJhdW
RpdHVzciIsIm1hdCI6MTY0MDY2NTk3MiwiYXBwcyI6ImRhdGFwdXNoIn0.z80h8i5
-0yygtVtY0lHRRfpIt8nf_t8v-hrrD8WrDu1XZ5bQ9q8Z5U4Tot9r7u6Hd-
m3DF_H5J9LqV_r1LvCXA' \
--header 'accept: application/json' \
--header 'content-type: application/json' \
--data-raw '{
  "vehiclenumber": "TG08T1024"
}'
```

For production –

<https://www.ulip.dpiit.gov.in/ulip/v1.0.0/TGVAHAN/01>

```
{  
  "vehiclenumber": "TG08T1024"  
}
```

Curl URL

```
curl --location  
'https://www.ulip.dpiit.gov.in/ulip/v1.0.0/TGVAHAN/01' \  
--header 'authorization: Bearer  
eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJ1bGwIiwiaWF0IjoxNzI3MTUxNjk3LCJh  
cHBzIjoizGF0YXB1c2gifQ.jnDTg3LoNNOS6ErRC4JH_i0bkhIVmv7TxWtis3KUqQ  
Wz_lq3Lr6mYqo94mzkz7mz-0Hb0lW40AUt-4p8TLT9Ig' \  
--header 'accept: application/json' \  
--header 'content-type: application/json' \  
--data '{  
  "vehiclenumber": "TG08T1024"  
}'
```

For obtaining an authentication token, refer [Section 1.5](#)

Response

To acknowledge the above request, ULIP will send the below response -

a) In case of an invalid format**1. Invalid vehicle number**

```
{  
  "response": null,  
  "error": "true",  
  "code": "400",  
  "message": "Data format failed OR wrong value entered a  
t: vehiclenumber. Format should follow ^[A-Z0-9]{5,11}$"  
}
```

b) In case of vehicle number does not exist in the TGVAHAN system

```
{
  "response": [
    {
      "response": "<NewDataSet>\r\n <errors>\r\n <isError>Y</isError>\r\n <errorMessage>No data found</errorMessage>\r\n </errors>\r\n <data>\r\n <Regn_No /\>\r\n <Owner_Name /\>\r\n <FatherName /\>\r\n <PresentAddress /\>\r\n <PermanentAddress /\>\r\n <Mobile_No /\>\r\n <RegistrationDate /\>\r\n <Chassis_No /\>\r\n <Engine_No /\>\r\n <Vehicle_Classdesc /\>\r\n <Maker_Name /\>\r\n <Model_Desc /\>\r\n <Body_Type /\>\r\n <Fuel /\>\r\n <Color /\>\r\n <FC_Validity /\>\r\n <rc_np_upto /\>\r\n <Tax_Validity /\>\r\n <rc_norms_desc /\>\r\n <FinancerName /\>\r\n <Insurance_CmpyName /\>\r\n <Insurance_No /\>\r\n <Insurance_Validity /\>\r\n <IssuePlace /\>\r\n <RcStatus /\>\r\n <Make_Yr /\>\r\n <ULW /\>\r\n <GVW /\>\r\n <NumberOfCylinder /\>\r\n <CubicCapacity /\>\r\n <SeatingCapacity /\>\r\n <SleeperCapacity /\>\r\n <StandCapacity /\>\r\n <WheelBase /\>\r\n <rc_np_issued_by /\>\r\n <OwnerSerialNo /\>\r\n <VehicleType /\>\r\n <PUCCVALIDFROM /\>\r\n <PUCCVALIDTO /\>\r\n <PERMITNO /\>\r\n <Permit_Validity /\>\r\n <PERMIT_TYPE /\>\r\n <TheftStatus /\>\r\n <RCValid_Upto /\>\r\n <Office_Code /\>\r\n <Vehicle_ClassID /\>\r\n <Approved_Date /\>\r\n <RCPurchaseDt /\>\r\n <NOCStay_Perd /\>\r\n <NOCIssueDt /\>\r\n <NOCTOOFFICE /\>\r\n </data>\r\n</NewDataSet>",
      "responseStatus": "SUCCESS",
      "message": null
    }
  ],
  "error": "false",
  "code": "200",
  "message": "Success"
}
```

c) In case the vehicle number exists

```
{
  "response": [
    {
      "response": "<NewDataSet>\r\n <errors>\r\n <isError>N</isError>\r\n <errorMessage /\>\r\n </errors>\r\n <data>\r\n <Regn_No>TG08T1024</Regn_No>\r\n <Owner_Name>S*I *A* E*T*R*S*S</Owner_Name>\r\n <FatherName>YALLANKI USHA</FatherName>\r\n <PresentAddress>PLOT NO.44 AND 45,FLAT NO. 401,4TH FLOOR,SRI VENKATA SAI COLONY,KOMPALLY</PresentAddress>\r\n <PermanentAddress>PLOT NO.44 AND 45,FLAT NO. 401,4TH FLOOR,SRI VENKATA SAI COLONY,KOMPALLY</PermanentAddress>\r\n <Mobile_No>9949648964</Mobile_No>\r\n <RegistrationDate>Apr 23 2024 4:07PM</RegistrationDate>\r\n <Chassis_No>MB1A5CHD7RPNA1586</Chassis_No>\r\n <Engine_No>RNPZ107043</Engine_No>\r\n <Vehicle_Classdesc>Goods Carriage</Vehicle_Classdesc>\r\n <Maker_Name>ASHOK LEYLAND LTD</Maker_Name>\r\n <Model_Desc>EA 1920/66 H CO FES BSVI-
```

```
PH2</Model_Desc>\r\n      <Body_Type>Closed</Body_Type>\r\n      <Fuel>DIESEL</Fuel>\r\n      <Color>NP</Color>\r\n      <FC_Validity>Apr 22 2026 2:45PM</FC_Validity>\r\n      <rc_np_upto />\r\n      <Tax_Validity>Sep 30 2024 12:00AM</Tax_Validity>\r\n      <rc_norms_desc />\r\n      <FinancerName>MAHINDRA &MAHINDRA FIN SER LTD</FinancerName>\r\n      <Insurance_CmpyName>CHOLAMANDALAM MS GENERAL INSURANCE COMPANY LTD.</Insurance_CmpyName>\r\n      <Insurance_No>123</Insurance_No>\r\n      <Insurance_Validity>Mar 14 2025 12:00AM</Insurance_Validity>\r\n      <IssuePlace>TG008</IssuePlace>\r\n      <RcStatus>Approved</RcStatus>\r\n      <Make_Yr>Feb 2 2024 12:00AM</Make_Yr>\r\n      <ULW>8720.00</ULW>\r\n      <GVW>18500.00</GVW>\r\n      <NumberOfCylinder />\r\n      <CubicCapacity>6600.00</CubicCapacity>\r\n      <SeatingCapacity>3</SeatingCapacity>\r\n      <SleeperCapacity />\r\n      <StandCapacity>0</StandCapacity>\r\n      <WheelBase>5760</WheelBase>\r\n      <rc_np_issued_by />\r\n      <OwnerSerialNo />\r\n      <VehicleType>T</VehicleType>\r\n      <PUCCVALIDFROM />\r\n      <PUCCVALIDTO />\r\n      <PERMITNO>TG008/2982/NPPUC/2024</PERMITNO>\r\n      <Permit_Validity>Apr 23 2029 4:11PM</Permit_Validity>\r\n      <PERMIT_TYPE>NP</PERMIT_TYPE>\r\n      <TheftStatus>N</TheftStatus>\r\n      <RCValid_Upto>Apr 22 2026 4:07PM</RCValid_Upto>\r\n      <Office_Code>TG008</Office_Code>\r\n      <Vehicle_ClassID>GCRT</Vehicle_ClassID>\r\n      <Approved_Date>Apr 23 2024 3:42PM</Approved_Date>\r\n      <RCPurchaseDt>Mar 15 2024 12:00AM</RCPurchaseDt>\r\n      <NOCStay_Perd />\r\n      <NOCIssueDt />\r\n      <NOCTOOFFICE />\r\n    </data>\r\n  </NewDataSet>",
    "responseStatus": "SUCCESS",
    "message": null
  }
],
"error": "false",
"code": "200",
"message": "Success"
}
```

Status Code 200: VRN data response

Status Code 400: Bad request (Invalid JSON syntax, invalid JSON key)

Status Code 401 or 403: Unauthenticated or Unauthorized

Status Code 500: Some internal server error occurred

Status Code 502: The server is not responding.

1.4 Data Transmission & Authentication Mechanism

All data exchange would be done over secure HTTP (HTTPS). Request from ULIP's system would be made in the TGVAHAN system through a data exchange URL that has the following form:

For Staging –

<https://www.ulipstaging.dpiit.gov.in/ulip/v1.0.0/TGVAHAN/01>

Body:

```
{"vehiclenumber": "TG08T1024"}
```

Access to above data exchange URL would be authorized by use of a security mechanism implemented by the ULIP system. It is as follows:

- First time access: It would require a username and password [which would be shared with user beforehand]. Once user hit with valid user id and password, ULIP system will return a basic authorization token to user, using that token user will communicate with ULIP system until token will not get expired.

Summary of HTTP Basic Authentication is as follows:

First time user hits ULIP secured API through username and password for access token as follow-

```
curl -X POST
'https://www.ulipstaging.dpiit.gov.in/ulip/v1.0.0/user/login'
-H 'accept: application/json'
-H 'content-type: application/json'
-data '{"username":"xxxxxx","password":"XXxxx@12345"} '
```

Authorization:**Bearer**

```
eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJpbmRLcmNvbnQiLCJpYXQiOiJlODU1NjYyMzcsImFwcHMiOiJkYXRhcHVzaCJ9.jb-23Dbkwpz-3ptsuuZofzH-ce7jd0xR5WLkFfBPO_Dzjop01dSx90amqsjsuAryLUPtUDiMX07oHG4L2AxFFQ
```

Upon successful authentication, the user would be logged into the ULIP system and get an access token. Though this token user can communicate with ULIP. This token has expiry time which represent user session. The session time out for the same to be fixed hour, generally it is for 30 minutes. If there is no request coming from user for 30 minutes, then session (token) will be expired, and user need to re authenticate.

Using The above token, user can request ULIP system for details as follow -

```
curl --location --
request POST 'https://www.ulipstaging.dpiit.gov.in/ulip/v1.0.0/TGVAHAN/01' \
--
header 'authorization: Bearer eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJlODU1NjYyMzcsImFwcHMiOiJkYXRhcHVzaCJ9.jb-23Dbkwpz-3ptsuuZofzH-ce7jd0xR5WLkFfBPO_Dzjop01dSx90amqsjsuAryLUPtUDiMX07oHG4L2AxFFQ' \
--header 'accept: application/json' \
--header 'content-type: application/json' \
--data-raw '{"vehiclenumber": "TG08T1024"}'
```

For Production –

```
curl --location 'https://www.ulip.dpiit.gov.in/ulip/v1.0.0/user/login' \  
--header 'Content-Type: application/json' \  
--header 'Accept: application/json' \  
--data-raw '{  
  "username": "xxxx",  
  "password": "xxxx@123"  
'
```

Authorization:**Bearer**

```
eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJ1bGwIiwiaWF0IjoxNzI3MTYxMDgwLCJhcHBzIjoizGF0YXB1c2gifQ.XhVZm7NiCD05-v9091WJQhYzdk1a0Shs_Ley0T4R2MX63swRyxHbHxh6bVXtzqWvWiSLwwNGRQIcqGhr49hpGg
```

Upon successful authentication, the user would be logged into the ULIP system and get an access token. Though this token user can communicate with ULIP. This token has expiry time which represent user session. The session time out for the same to be fixed hour, generally it is for 30 minutes. If there is no request coming from user for 30 minutes, then session (token) will be expired, and user need to re authenticate.

Using The above token, user can request ULIP system for vehicle details as follow -

<https://www.ulip.dpiit.gov.in/ulip/v1.0.0/TGVAHAN/01>

```
curl --location 'https://www.ulip.dpiit.gov.in/ulip/v1.0.0/TGVAHAN/01' \  
--header 'authorization: Bearer  
eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJ1bGwIiwiaWF0IjoxNzI3MTYxMDgwLCJhcHBzIjoizGF0YXB1c2gifQ.XhVZm7NiCD05-v9091WJQhYzdk1a0Shs_Ley0T4R2MX63swRyxHbHxh6bVXtzqWvWiSLwwNGRQIcqGhr49hpGg' \  
--header 'accept: application/json' \  
--header 'content-type: application/json' \  
--data '{"vehiclenumber": "TG08T1024"}'
```